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Switching Rules and Mismatch Dynamics in Context-Dependent Encoding

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Abstract
Sensory systems must adapt their encoding strategies when environmental statistics change over time. In changing environments, this requires not only context inference but also a rule governing when the system switches between context-specific encodings. Here, I study how switching rules affect performance in a simplified one-neuron model of context-dependent encoding. I compare two forms of environmental change, mean switching and variance switching, and evaluate instant switching, hysteresis, and refractory lockout switching rules while holding the set of context-specific encoding functions fixed. Performance and mismatch dynamics were quantified using global mean squared error (MSE), context misclassification rate, mean mismatch duration, and mismatched MSE. Within each environment, switching rules had little effect on mismatched MSE itself, but substantially affected mismatch frequency and duration. In the mean-switching environment, these changes substantially altered overall performance, whereas in the variance-switching environment they had weaker effects on global MSE. These results show that, when context must be inferred online, adaptive performance depends not only on the available codes but also on the dynamics governing transitions between them.

Significance Statement

Because environmental conditions shape the statistical structure of sensory input, a sensory system may need to change how it converts input into neural signals under different conditions. This study tests whether doing well under changing conditions depends only on having suitable methods of converting sensory input into neural signals, or also on when the system switches between them. By holding those encoding methods fixed, the study isolates the effect of switching rules alone. In a simple one-neuron model, different switching rules changed how often the system used a method less suited to the current environmental condition and how long that mismatch lasted. Those timing differences changed how much error accumulated from mismatches.

Introduction

A fundamental challenge for sensory systems is to represent environmental information efficiently using limited coding resources. One influential proposal is the efficient coding hypothesis, which holds that sensory systems are adapted to produce efficient representations of their environment under biological constraints (1). Classic quantitative work has shown how neural response properties can be efficiently matched to stimulus distributions to optimize information transmission (2). However, many such treatments assume a stationary environment, in which input statistics remain fixed over time.

Młynarski and Hermundstad extended this perspective to dynamic environments, asking how sensory systems should adapt their codes when context changes while encoding resources remain limited (3). This extension is necessary because natural environments are not stationary, and sensory systems must continually adapt to changing input statistics (4, 5). In their normative framework, a sensory system must infer context and adapt its encoding accordingly. Because context estimates are imperfect, the selected code can be matched or mismatched to the true context. Global error therefore depends not only on coding quality within a context but also on how often mismatch occurs, how long it persists, and how costly it is when it occurs. A code that is locally optimal for the inferred context need not minimize long-run error when inference is imperfect.

The central question in this study is how switching rules alter performance when context must be inferred online. Rather than assuming immediate adoption of the inferred context, I compare an instant-switching baseline with two more conservative switching rules: hysteresis and refractory lockout. Under hysteresis, the system switches only when posterior belief in the alternative context exceeds a threshold. Under refractory lockout, once the encoding function changes, further changes are temporarily disallowed. The overall adaptive-coding framework in this study follows Młynarski and Hermundstad; however, the context-specific encoding functions were held fixed while the switching rules were varied. By holding the context-specific encoding functions fixed and varying only the switching rule, this study isolates how switching dynamics alone shape mismatch structure and global performance. I study this question in two controlled environments, one in which contexts differ in stimulus mean and one in which they differ in stimulus variance.

Results

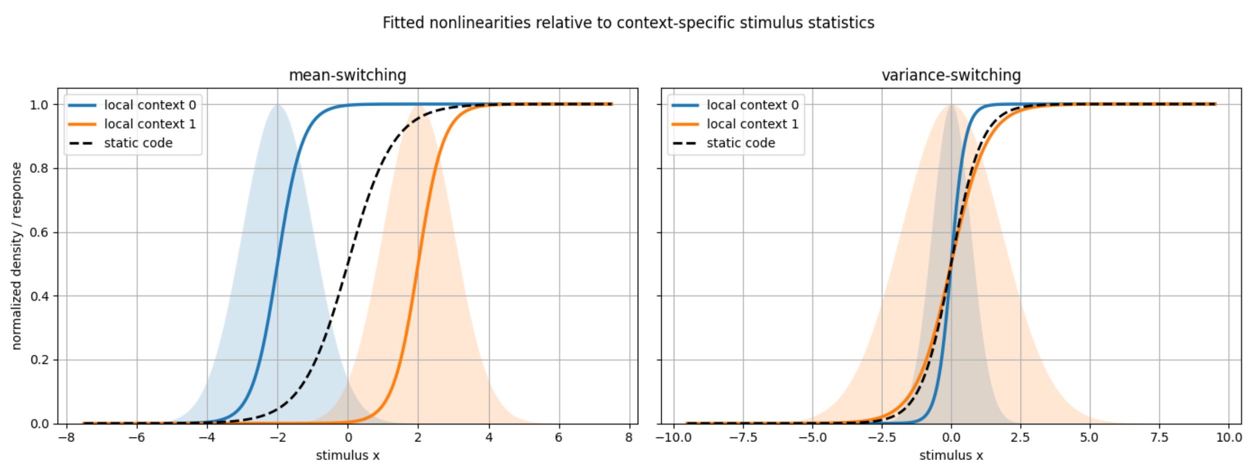

Figure 1. Optimized nonlinear response functions in the mean- and variance-switching environments. Shaded curves show context-specific stimulus distributions, solid curves show local codes, and the dashed curve shows the static code optimized for the mixture distribution.

I first examined the optimized encoding functions in the two environments (Figure 1). In the mean-switching environment, the local nonlinearities are primarily shifted along the stimulus axis, whereas in the variance-switching environment they overlap more in position and differ mainly in slope. This distinction provides a structural difference between the two environments that helps frame the performance comparisons below.

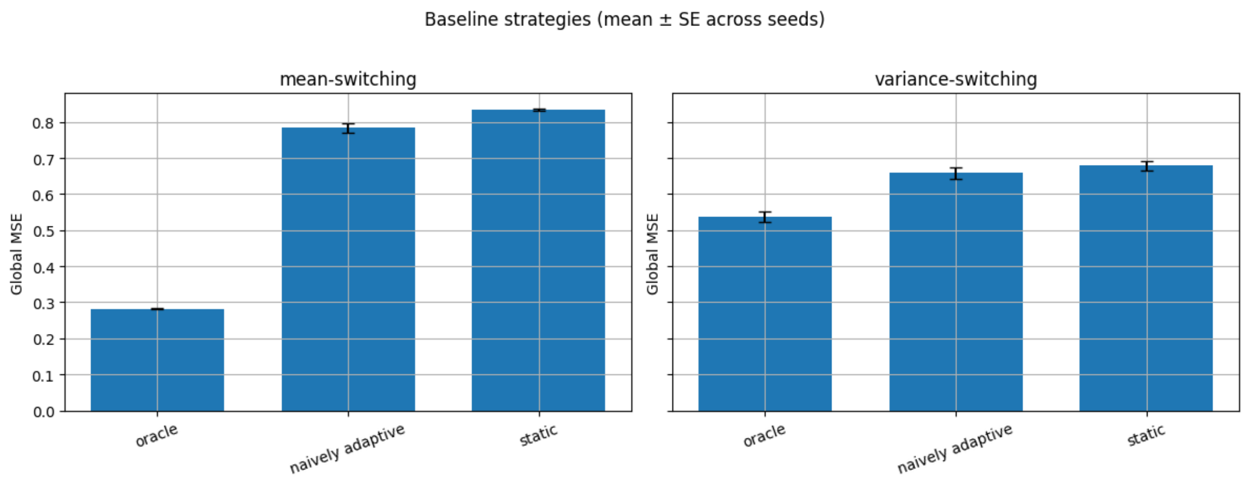

Figure 2. Baseline comparison of oracle, naively adaptive, and static strategies. Bars show global MSE in the mean-switching and variance-switching environments (mean ± SE across seeds).

To establish a performance baseline, I compared static, naively adaptive, and oracle strategies (Figure 2). In both environments, oracle encoding achieves the lowest global MSE, since it uses the true context and never mismatches. Static encoding yields higher error because it does not adapt to context changes. The naively adaptive strategy lies between these extremes, recovering part of the benefit of context-specific coding without eliminating mismatch. The gap between oracle and the other strategies is much larger in the mean-switching environment than in the variance-switching environment. Under these parameter settings, context-specific adaptation yields a larger performance gain when contexts differ in mean than when they differ only in variance.

I next examined how switching rules reshape mismatch structure and overall performance (Figure 3). In both environments, mismatched MSE changed little across switching rules, indicating that they do not substantially alter the cost of mismatch. Their main effect was on mismatch frequency and duration. In the mean-switching environment, instant switching and hysteresis produced similar global MSE and context misclassification rates, although hysteresis yielded longer mismatch episodes. Refractory lockout increased both mismatch rate and mismatch duration and correspondingly produced the highest global MSE. In the variance-switching environment, switching rules still altered mismatch structure, especially mismatch duration, but global MSE changed only weakly across rules. Hysteresis produced the longest mismatch episodes in this environment. Thus, switching rules primarily reshaped the temporal structure of mismatch, and the impact of those changes on global performance depended on the environment.

As an additional extension, I examined how stronger context separation changes the balance between inference and mismatch cost. In both environments, stronger separation reduced context misclassification and shortened mismatch episodes, but increased mismatched MSE.

Global MSE increased in both environments, but the increase was much larger in the mean-switching environment and comparatively small in the variance-switching environment. This trade-off was substantially stronger in the mean-switching environment than in the variance-switching environment. The full metric breakdown for these manipulations is shown in the SI Appendix, Figs. S1 and S2.

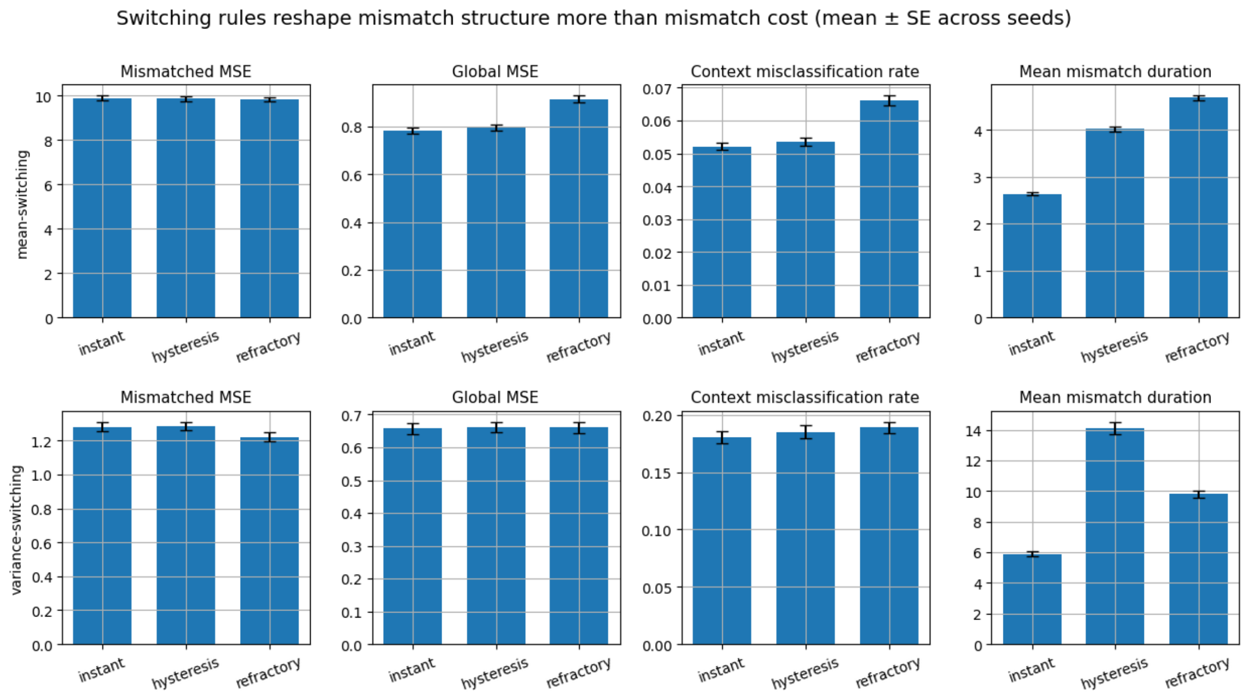

Figure 3. Switching rules reshape mismatch structure more than mismatch cost. Bars show mismatched MSE, global MSE, context misclassification rate, and mean mismatch duration for instant switching, hysteresis, and refractory lockout in the mean-switching and variance-switching environments (mean ± SE across seeds).

Discussion

The main result of this study is that, once the encoding functions were held fixed, switching rules affected global performance mainly by reshaping mismatch over time rather than by substantially changing mismatch cost. Across switching-rule comparisons in both environments, mismatched MSE changed little, whereas context misclassification rate and mean mismatch duration changed more strongly. Global MSE differences across switching rules therefore tracked mismatch frequency and duration more closely than mismatched MSE. This pattern was clearest in the mean-switching environment, where refractory lockout produced the largest increases in mismatch rate, mismatch duration, and overall error.

These results suggest that switching should be treated as part of the adaptive strategy rather than as a secondary control mechanism, because once the encoding functions were held fixed, overall performance still depended on how transitions between them were regulated. The separation analysis reinforces this point by showing that stronger context separation reduced misclassification and mismatch duration but increased mismatched MSE, so the effect of switching depends on how environmental structure balances mismatch dynamics against mismatch cost. Consistent with this trade-off, switching-rule effects were stronger in the mean-switching environment than in the variance-switching environment, indicating that the impact of switching depends on environmental structure.

Although the switching rules studied here were heuristic, they can be interpreted as simple abstractions of history dependence and temporary stabilization mechanisms that have biological parallels (6–8). The model remains highly simplified, using a single neuron, binary contexts, Gaussian stimulus statistics, and a small set of hand-designed switching rules. This simplification helps isolate switching dynamics, but leaves many other adaptive settings unexplored. Future work should test whether the same mechanism persists in population codes, richer context spaces, and more biologically grounded switching rules.

Materials and Methods

I considered two environments. In the mean-switching environment, the two contexts differed in mean while sharing the same variance:

$$X_t \mid C_t = 0 \sim \mathcal{N}(\mu_0, \sigma^2), \qquad X_t \mid C_t = 1 \sim \mathcal{N}(\mu_1, \sigma^2), \qquad \mu_0 \neq \mu_1$$

with default parameters $\mu_0 = -2$, $\mu_1 = 2$, and $\sigma = 1$. In the variance-switching environment, the two contexts shared the same mean but differed in variance:

$$X_t \mid C_t = 0 \sim \mathcal{N}(\mu, \sigma_0^2), \qquad X_t \mid C_t = 1 \sim \mathcal{N}(\mu, \sigma_1^2), \qquad \sigma_0 \neq \sigma_1$$

with default parameters $\mu = 0$, $\sigma_0 = 0.7$, and $\sigma_1 = 1.8$. The latent context followed a two-state Markov chain with switch probability $p_{\text{switch}} = 0.01$:

$$P(C_t \neq C_{t-1}) = p_{\text{switch}}$$

The encoder was modeled as a logistic nonlinearity with adaptable slope a, offset b, and fixed maximum response $R_{\max} = 1$:

$$f(x; a, b) = \frac{R_{\max}}{1 + \exp[-a(x - b)]}$$

The noisy observed response was modeled as:

$$R_t \mid X_t = x \sim \mathcal{N}(f(x; a, b), \sigma_r^2),$$

with $\sigma_r = 0.18$. For a given response r, the stimulus estimate was the posterior mean $\hat{x}(r) = \mathbb{E}[X \mid R = r]$. Posterior quantities were computed numerically on discretized stimulus and response grids. For each context, I defined a local code as the parameter pair (a, b) that minimized expected squared decoding error under that context's stimulus distribution. I also computed a static code by minimizing the same objective under the mixture distribution across contexts. These encoding functions were then held fixed throughout the simulations. At each time step, the system updated a posterior belief over contexts from the previous belief, the context transition matrix, and the response likelihood. Full recursive update equations are given in the SI Appendix. The inferred context was taken to be the maximum a posteriori (MAP) estimate. This inferred context selected the candidate local code, subject to the switching rule. Under instant switching, the system immediately adopted the encoding function associated with the inferred context. Under hysteresis, the system switched encoding only when posterior belief in the alternative context exceeded 0.65 as a default conservative switching criterion. Under refractory lockout, once a switch occurred, no further switch was allowed for 5 time steps. The naively adaptive strategy used the MAP-inferred context with instant switching.

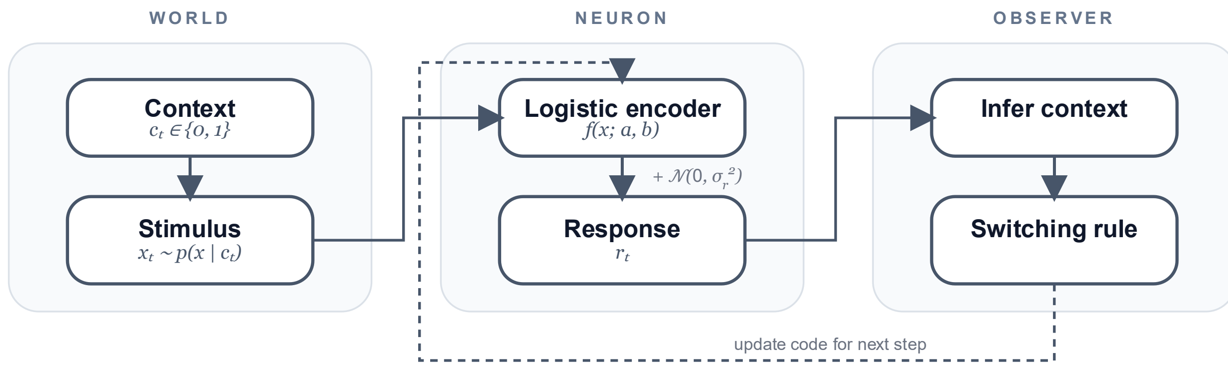

Figure 4. Simulation framework**.** A latent context determines the stimulus distribution; the resulting noisy response is used to update belief about context, and the updated belief together with the switching rule determines which encoding function is used at the next time step.

Figure 4 outlines the simulation framework. I compared the switching rules against two baselines: an oracle strategy, which used the true context, and a static strategy, which always used the mixture-optimized code. Simulations were run for 5000 time steps with the first 500 discarded as burn-in, and results were averaged over 20 random seeds. Performance was evaluated using four metrics. Here, $\hat{C}_t$ denotes the context associated with the encoding function actually used at time t after applying the switching rule. Global mean squared error (global MSE) was defined as Global MSE $=\frac{1}{T}\sum_{t=1}^{T}(X_t - \hat{X}_t)^2$. Context misclassification rate was defined as $\frac{1}{T}\sum_{t=1}^{T} 1[\hat{C}_t \neq C_t]$. Mean mismatch duration was defined as the average length of consecutive mismatch episodes, $\frac{1}{K}\sum_{k=1}^{K} d_k$, where d_k is the duration of the k-th mismatch episode. Finally, mismatched MSE was defined as Mismatched MSE $= \frac{1}{|M|}\sum_{t\in M}(X_t - \hat{X}_t)^2$, where $M = \{t: \hat{C}_t \neq C_t\}$. Together, these quantities separate how often mismatch occurs, how long it lasts, and how costly it is when it occurs.

Data, materials, and code availability

All simulations were implemented in Python for this project. The theoretical framing follows Młynarski and Hermundstad (3), but no external codebase or dataset was directly reused.

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